

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA1001	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
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Date:	01/10/2026	Duration:	60 minutes

Code of Conduct

I Hari Prakash. N / 192572038 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Hari Prakash. N

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

CASE STUDY ANALYSIS

Food Waste Management and Donation Platform

Problem Statement: Food waste is generated by households, restaurants, hotels, supermarkets, caterers, events, and other food-serving organizations even when some surplus food may still be suitable for consumption. At the same time, charitable organizations and communities may have difficulty locating and receiving available food in a timely manner. The proposed Food Waste Management and Donation Platform is a software-based system that connects food donors with verified recipient organizations, coordinates donation requests, supports pickup or delivery scheduling, and maintains records for transparency and reporting.

1. Understand the Problem Statement

1.1 Problem Explanation

- A lot of good and usable food is wasted every day from homes, hotels, restaurants, canteens, and events.

- At the same time, many people and organizations need food but do not know where surplus food is available.
- Currently, food donations are often managed through phone calls, WhatsApp groups, or personal contacts.
- It is difficult to know what food is available, how much is available, where it is located, and when it should be collected.
- Sometimes, the food may become unsuitable for use because there is a delay in finding the right recipient.
- Donors may also not know which organization or person can properly receive and distribute the food.
- Therefore, there is a need for a simple and organized way to connect food donors with suitable recipients.
- The proposed Food Waste Management and Donation Platform can help donors provide details about surplus food and help recipients find available food.
- The platform can also help track the food from donation to collection/delivery, reducing unnecessary food waste.

1.2 Objectives of the Proposed System

- Create a centralized platform for registering and managing surplus food donations.
- Connect food donors with verified NGOs, charities, community kitchens, shelters, or other eligible recipient organizations.
- Reduce delays by providing timely notifications about newly available donations and collection deadlines.
- Improve coordination of pickup and delivery by recording locations, time windows, quantities, and responsible parties.
- Maintain a transparent status history for each donation from listing through acceptance and completion.
- Provide reports and basic analytics on donations, quantities distributed, completed pickups, cancellations, and recurring donation patterns.
- Support responsible handling of food by capturing relevant information such as food type, quantity, preparation time, and use-by or collection deadline.

1.3 Scope of the Problem

The scope includes donor registration, recipient/NGO registration and verification, food donation listing, search and matching, donation acceptance, pickup/delivery coordination, notifications, status tracking, feedback, reporting, and administrative monitoring. The platform can be designed as a web or mobile-accessible application with a centralized database.

The scope does not require the platform itself to prepare food, physically transport donations, guarantee food safety, or replace the legal and organizational responsibilities of donors and recipient organizations. Instead, it provides the information and coordination infrastructure through which participating stakeholders can manage donations more systematically.

2. Identify Key Stakeholders and Challenges

2.1 Stakeholders, Roles and Responsibilities

Stakeholder	Role / Responsibility	Major Challenges
Food Donors	Restaurants, hotels, supermarkets, caterers, households and event organizers list surplus food and provide accurate quantity, type, location and collection details.	Time-sensitive surplus; uncertainty about recipient availability; manual coordination; concerns about accountability.
Recipient Organizations	NGOs, shelters, community kitchens, charities and approved groups search for suitable donations, accept them and arrange collection.	Limited transport/capacity; changing demand; difficulty discovering donations quickly.
Beneficiaries	People or communities ultimately receive donated food through participating organizations.	Need for timely access; dietary or quantity requirements; dependence on recipient organizations.
Platform Administrator	Manages users, verifies organizations, monitors transactions, handles complaints and maintains platform rules.	Fraudulent accounts; inaccurate listings; dispute handling; data protection; system availability.
Delivery / Pickup Partners	Volunteers, authorized drivers or logistics partners collect and transport accepted donations when applicable.	Scheduling conflicts; route and time constraints; accurate pickup information required.
Development / Maintenance Team	Designs, develops, tests, deploys, secures and maintains the platform.	Changing requirements; integration issues; security vulnerabilities; maintenance and scalability.
Authorities / Regulators, where applicable	Provide applicable guidance concerning food donation, hygiene, transportation, records or community programs.	Need for compliance, traceability and adherence to applicable requirements.

2.2 Major Challenges and Issues

- Time sensitivity: surplus food may have a limited safe-use window, so slow matching or delayed pickup can cause a donation to be lost.
- Information accuracy: incorrect quantities, locations, food descriptions, or collection deadlines can create failed pickups and wasted resources.
- Trust and verification: the platform must distinguish genuine donors and recipient organizations from fraudulent or unsuitable accounts.
- Logistics coordination: donors and recipients may have different schedules, transport availability, capacities, and geographic constraints.
- Food suitability: the platform can record relevant information, but responsibility for appropriate handling and acceptance remains with participating parties.

- Uneven supply and demand: some areas may have more surplus food while other areas have greater demand, requiring efficient matching.
- Privacy and security: contact details, addresses, account credentials, transaction records, and organizational information must be protected.
- User adoption: organizations may continue using manual methods unless the platform is simple, reliable, and clearly useful.

3. Analyze the Current Approach

3.1 Existing System / Current Process

In a typical manual process, a donor identifies surplus food and contacts a known NGO, charity, community group, or volunteer through phone calls or messaging applications. The receiving organization checks whether it has capacity and transport. If accepted, both parties agree on a pickup time and location. The donation may then be collected and distributed. Records, if maintained, are often kept in spreadsheets, notebooks, messages, or separate organizational systems.

Some organizations may already use social media or messaging groups to announce available food. These channels can improve communication compared with completely offline coordination, but they are not necessarily designed for structured donation matching, transaction status tracking, verification, or consolidated reporting.

3.2 Strengths of the Current Approach

- Simple to start because it can use existing communication channels and personal networks.
- Direct communication allows donors and recipients to clarify special requirements quickly.
- Local organizations may already have established trust and knowledge of nearby communities.
- Manual coordination can work for a small number of recurring donors and recipients.

3.3 Limitations and Gaps

- No single searchable inventory of currently available surplus food.
- Limited ability to automatically match donations with recipient requirements, location, capacity and time constraints.
- Messages and calls can be missed, causing time-sensitive donations to expire before collection.
- Donation status is difficult to track consistently across multiple people and organizations.
- Verification of users and organizations may be informal or inconsistent.
- Historical records and reports may require manual consolidation from different sources.
- Limited visibility into recurring patterns such as available food types, demand levels, completed donations and cancellation causes.
- Scalability is difficult when the number of donors, recipients and daily donation listings increases.

3.4 Key Gap to be Addressed

The central gap is not simply the absence of communication; it is the absence of an integrated workflow that combines verified participation, structured donation information, time-sensitive matching,

coordination, status tracking, and reporting. The proposed platform addresses this gap by making the donation lifecycle visible and manageable through a common system.

4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Proposed Software Solution

The proposed solution is a centralized Food Waste Management and Donation Platform with separate but connected interfaces for donors, recipient organizations, delivery or pickup partners, and administrators. A donor creates a donation listing containing food details, quantity, location, availability window and relevant handling information. The matching component identifies recipient organizations that meet configurable criteria such as location, capacity, food category and availability. The recipient can accept or decline the offer. For accepted donations, the system records a pickup or delivery schedule and updates the status through stages such as Listed, Matched, Accepted, Scheduled, Collected/Delivered, Completed, Cancelled, or Expired.

The platform should use role-based access so that each stakeholder sees only the functions required for their role. A centralized database maintains user, organization, donation, matching, scheduling, notification, feedback and audit information. Notifications can be triggered for new matches, acceptance, upcoming pickup windows, cancellations and expiry.

4.2 Major Functional Modules

Module	Main Functions
1. User Registration and Authentication	Registration, login, password management, role assignment and account verification.
2. Donor Management	Donor profile, organization details, donation history and recurring donation information.
3. Recipient Organization Management	Organization profile, verification status, food preferences, capacity, service area and availability.
4. Food Donation Listing	Create, edit and close listings with food type, quantity, location, preparation/availability time and deadline.
5. Donation Matching	Identify compatible recipients using configurable criteria such as distance, capacity, food category and time window.
6. Acceptance and Scheduling	Accept or decline donation, assign pickup or delivery time, and record responsible person or logistics partner.
7. Notification and Alerts	Send relevant alerts for new donations, matches, acceptance, schedule changes and approaching deadlines.
8. Donation Tracking	Maintain current status and timestamp history for every donation.
9. Feedback and Issue Reporting	Capture completion feedback, failed pickup reasons, complaints and resolution status.

10. Reporting and Administration	Dashboard, donation statistics, user verification, audit records, exception monitoring and report generation.
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4.3 Functional Requirements

- The system shall allow users to register and authenticate according to their roles.
- The system shall allow authorized donors to create, update and close food donation listings.
- The system shall store donation details including type, quantity, location, availability window and relevant deadline information.
- The system shall allow verified recipient organizations to view and respond to suitable donation offers.
- The system shall support matching based on configurable criteria such as location, capacity, food category and time availability.
- The system shall allow accepted donations to be scheduled for pickup or delivery.
- The system shall update and display donation status throughout the donation lifecycle.
- The system shall send notifications for important events such as matching, acceptance, schedule changes and approaching deadlines.
- The system shall maintain donation history and generate administrative reports.
- The system shall provide mechanisms for feedback, cancellation, issue reporting and administrative resolution.

4.4 Non-Functional Requirements

- Usability: interfaces should be simple enough for users with different levels of technical experience.
- Reliability: donation records and status changes should be stored consistently, with appropriate error handling and recovery.
- Performance: common operations such as listing, searching and viewing donations should respond within an acceptable time under expected load.
- Security: authentication, authorization, secure password handling, input validation, encryption where appropriate, and audit logging should be applied.
- Scalability: the architecture should support growth in users, donation records and geographic coverage without major redesign.
- Maintainability: modules should be separated and documented so changes to one feature have limited impact on others.
- Availability: the service should be designed to remain accessible during expected operating periods, with backup and recovery procedures.
- Privacy: only necessary personal and organizational information should be collected, stored and exposed to other users.

4.5 Application of Software Engineering Principles

Principle	How It Is Applied
Modularity	Separate authentication, donation management, matching, scheduling, notifications, reporting and

	administration into distinct modules with clear interfaces.
Scalability	Use layered or service-oriented architecture, indexed database structures and scalable hosting so additional users and locations can be supported.
Maintainability	Use consistent coding standards, documentation, version control, automated testing and separation of business logic from presentation.
Reliability	Validate inputs, use transactional database operations where required, maintain backups, log important events and provide controlled error handling.
Usability	Provide role-specific dashboards, simple forms, clear status labels, search/filter options and meaningful notifications.
Security	Apply role-based authorization, secure authentication, data protection, input validation, audit logging and least-privilege access.
Reusability	Develop common components such as authentication, notification and reporting services so they can be reused across modules.
Testability	Define requirements and acceptance criteria for each module and use unit, integration, system and user-acceptance testing.

5. Justify the Chosen Methodology / Framework

5.1 Selected Methodology: Agile with Scrum

Agile with Scrum is suitable for this case study because the platform contains several independent modules and its requirements may evolve after feedback from donors, recipient organizations and administrators. Instead of attempting to finalize every detail before development, the project can be organized into short iterations in which a small set of usable features is designed, developed, tested and reviewed.

A possible product backlog can include user authentication, donor registration, recipient verification, donation listing, matching, scheduling, notifications, tracking, reporting and administration. The team can prioritize the backlog according to stakeholder value and implementation dependencies. Each sprint can produce a demonstrable increment, allowing stakeholders to identify usability or workflow problems early.

5.2 Why Agile/Scrum Fits the Proposed System

- Requirements can be refined through regular feedback from real users such as donors and recipient organizations.
- The system can be delivered incrementally, beginning with core donation listing and matching before adding advanced reporting and analytics.

- Frequent testing reduces the risk of discovering major defects only at the end of development.
- Short iterations support changing operational requirements such as notification rules, matching criteria and administrative workflows.
- Scrum roles, sprint planning, reviews and retrospectives provide a structured mechanism for communication and continuous improvement.

5.3 Support for Development and Maintenance

During development, the product backlog can be maintained by a Product Owner, the development team can implement sprint goals, and the Scrum Master can support the process and remove workflow impediments. Sprint reviews provide an opportunity to inspect the increment with stakeholders, while retrospectives help the team improve its development process. After deployment, new requirements, defects, security updates and enhancement requests can be converted into backlog items and prioritized for future iterations. This provides a continuous maintenance path rather than treating maintenance as a separate one-time activity.

6. Discuss Expected Outcomes and Limitations

6.1 Expected Outcomes and Benefits

- Improved visibility of available surplus food through structured donation listings.
- Faster communication between donors and recipient organizations through matching and notifications.
- Better coordination of collection and delivery through recorded schedules and status updates.
- Reduced dependence on scattered phone calls, messages and manual registers.
- Improved traceability through a history of donation creation, acceptance, scheduling and completion.
- Administrative reports that help participating organizations understand donation volume, completion rates and recurring operational issues.
- Potential improvement in the ability of recipient organizations to discover donations that match their capacity and service area.

6.2 How the Solution Addresses Identified Challenges

The time-sensitivity challenge is addressed through availability windows, deadlines and notifications. Information accuracy is improved by structured listing fields and validation. Trust is addressed through account verification and role-based access. Logistics challenges are reduced by recording location, capacity and collection schedules. Coordination gaps are reduced by a single status workflow that makes it clear whether a donation is listed, accepted, scheduled, collected, completed, cancelled or expired. Reporting reduces the effort required to consolidate historical records.

6.3 Risks, Implementation Challenges and Limitations

Risk / Challenge	Potential Effect	Mitigation
Incorrect or incomplete donation information	May result in unsuitable matches or failed pickups.	Use mandatory fields, validation, clear guidance and confirmation before submission.
Food safety / suitability concerns	The platform cannot independently guarantee that donated food is safe or suitable.	Capture relevant information and clearly assign responsibility to participating organizations

		according to applicable procedures and requirements.
Low user adoption	Donors or recipient organizations may continue using informal communication channels.	Keep workflows simple, provide useful notifications/reports and gather user feedback during pilots.
Missed pickups	Transport or scheduling problems may cause accepted donations to expire.	Provide reminders, cancellation/reassignment options and pickup status tracking.
Fraudulent or unsuitable accounts	Fake accounts may misuse the platform or create unreliable listings.	Use verification, role-based permissions, reporting mechanisms and administrative review.
Privacy and cybersecurity incidents	Unauthorized access could expose account or organizational information.	Apply secure authentication, authorization, encryption where appropriate, logging, backups and security testing.
Uneven regional coverage	The platform may be less useful where there are few donors or recipients.	Support gradual geographic expansion and partnerships with local organizations.
Scalability and availability	Growth in transactions can affect performance if architecture is not designed for scale.	Use modular architecture, monitoring, database optimization and scalable infrastructure.

6.4 Future Enhancements

- Mobile application with location-aware donation discovery and pickup coordination.
- Intelligent matching that considers historical acceptance patterns, travel distance, capacity and time constraints.
- Route optimization for delivery or volunteer pickup scheduling.
- Analytics dashboards for identifying high-waste locations, recurring donation times and unmet recipient demand.
- Integration with notification services and optional mapping/location services.
- QR or digital confirmation for handover and completion tracking.
- Multilingual interfaces to support broader community participation.
- Integration with organization-level inventory or point-of-sale systems where appropriate.
- Enhanced accessibility features for users with different abilities and devices.

Overall Case Study Conclusion

The Food Waste Management and Donation Platform addresses a coordination problem by connecting surplus-food donors with verified recipient organizations through a structured digital workflow. The proposed system covers the complete donation lifecycle from listing and matching to scheduling, collection or delivery, completion and reporting. Its design applies software engineering principles

including modularity, scalability, maintainability, reliability, usability and security. Agile with Scrum provides an iterative approach for developing the platform because stakeholder feedback and operational requirements can evolve as the system is evaluated. The proposed solution can improve visibility, coordination and traceability while recognizing that food handling, logistics, user participation, applicable requirements and data security remain important implementation considerations.

Rubric Alignment Summary

This analysis is structured to address the assessment criteria directly: Section 1 provides complete problem understanding, objectives and scope; Section 2 identifies stakeholders, roles and challenges; Section 3 analyzes the existing process, strengths, limitations and gaps; Section 4 presents the proposed software solution, requirements and software engineering principles; Section 5 selects and justifies an SDLC framework; and Section 6 covers outcomes, challenge resolution, risks, limitations and future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability,	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.

	maintainability, security, etc.).			
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25